HEAT EXCHANGER ANALYSIS

LUKE KRICK, AUBREY WRIGHT, JACK DURHAM, RACHEL PHILLIPS, KATIE BRENEMAN

TEAM CONTRIBUTIONS

- Aubrey Wright
  - Completed Geometric Calculations
  - Completed Heat Exchanger Analysis
- Luke Krick
  - Completed ANSYS Fluent Simulation
  - Completed Mesh Iterations
- Katie Breneman
  - Wrote the report
  - Created/ Organized Presentation
- Rachel Phillips
  - Wrote the report
  - Created/ Organized Presentation
- Jack Durham
  - Completed SolidWorks model and drawing
  - Ethical considerations

INTRODUCTION

Problem/Client's Deliverables

- Design a shell and tube heat exchanger
- One shell pass and one tube pass with N tubes
- 1 meter < Length < 10 meters
- Determine the ideal amount of tubes

	$\dot{m}$(kg/s)	$T_{m,i}$ (°C)	$T_{m,o}$ (°C)
Hot Water	28	90	-
Cold Water	27	34	60

Effectiveness-NTU Method

- Used when outlet temperatures are not given
- Effectiveness: $\in = \frac{q}{q_{max}}$
- Maximum Heat Rate: $q_{max} = C_{min}(T_{h,i} - T_{c,i})$
- NTU = $\frac{UA}{C_{min}}$

VALUE OVERVIEW

Dimension	Value	Units
Outlet Hot Temperature, $T_{h.o}$	338.02	K
Heat Rate, q	6.32	MW
Effectiveness, ε	0.464	—
NTU_{cf}	0.852	—
Pressure Loss per pipe, Δp	228.43	Pa
Pumping Power, $\dot{W}$	239.6	W

Dimension	Value	Units
Number of tubes, N	38	—
Length, L	9	m
Shell Diameter, D_o	1.20	m
Inner Diameter, D_i	0.0500	m
Cross-Sectional Area of Inner Pipes, $A_{c,i}$	$1.96(10^{-3})$	m^2
Cross-Sectional Area of Outer Pipe, $A_{c,o}$	1.13	m^2
Surface Area of Inner Pipes, $A_{s,i}$	58.00	m^2
Surface Area of Outer Pipe, $A_{s,o}$	33.93	m^2

```
Aci = (pi*D.^2) / 4 ;%m^2 Cross sectional area of small tubes
Aco = (pi*Do.^2) / 4 ;%m^2 cross sectional area of shell
Asi = n*pi*D*L; %m^2 Surface area inside tubes
Aso = n*pi*Do*L; %m^2 Surface area Annulus, won't need this bc its insulated
% Shellsa=
% Shellcs=
D_2 = 1.05*D;   % this is an imaginary diameter around the pipe, factor in the gap between the pipes

P = pi * (D + D_2) ;%m one pipe and its imaginary space
Ac_2 = n*pi/4*(D^2 + D_2^2); %m^2 cross-sec area of imaginary gap, total cross sec gap between all inner tubes

%velocity through pipe
Vi = mfc ./ (n*Aci*rowaterc); %m/s

%reynolds thru pipe
Rei = 4*mfc ./ (pi *D *muc);
Reo=4*mfh ./ (pi*Do*muh);

%solve for h value inside
Nui = 0.023 .* Rei .^(4/5) .* prc ^ (0.4);
hi = Nui .* kc ./ D; %W/m^2*K

%Hydraulic diameter
D_hyd = 4*Ac_2/(pi*n); %m

%solve for h value outside
Nuo = 0.023 .* Reo .^(4/5) .* prh ^ (0.3)
ho = Nuo .* kh ./ D_hyd; %W/m^2*K

%Conduction resistance - take into account
Rcond= (log(Do/D)/ (2*pi*kcu*L)); %ohm

% calculate U based on both h's and Conduction resistance
Rtot = (1 ./ (hi*Asi)) + (1 ./ (ho*Aso)) + (Rcond); %ohm
U=(1/ (Rtot*Asi)) ;%W/m^2*K
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EFFECTIVENESS-NTU METHOD CALCULATIONS

OUTLET HOT TEMPERATURE AND HEAT RATE

- All fluid properties found with $T_{c,o}$ = 333 K
- Equation 11.7b used, q_c = 2934.46 kW
- $q_c = q_h$
- Initial hot outlet temperature $T_{h,o}$ was chosen to find specific heat for the hot side $c_{p,h}$. Then used to find corresponding $T_{h,o}$
- Iterations performed via MATLAB to give $\boldsymbol{T_{h,o}}$ = **338.02 K**
- Equation 11.36 used to find heat capacity ratio C_r = 0.9607
- Equation 11.18 gives $\mathbf{q_{max}}$ **= 6.32 MW**

$$q = \dot{m}_c c_{p,c}(T_{c,o} - T_{c,i}) \qquad (11.7b)$$

$$\frac{C_{\min}}{C_{\max}} = \frac{\dot{m}_h c_{p,h}}{\dot{m}_c c_{p,c}} = \frac{T_{c,o} - T_{c,i}}{T_{h,i} - T_{h,o}} \qquad (11.26)$$

$$q_{\max} = C_{\min}(T_{h,i} - T_{c,i}) \qquad (11.18)$$

EFFECTIVENESS AND NTU

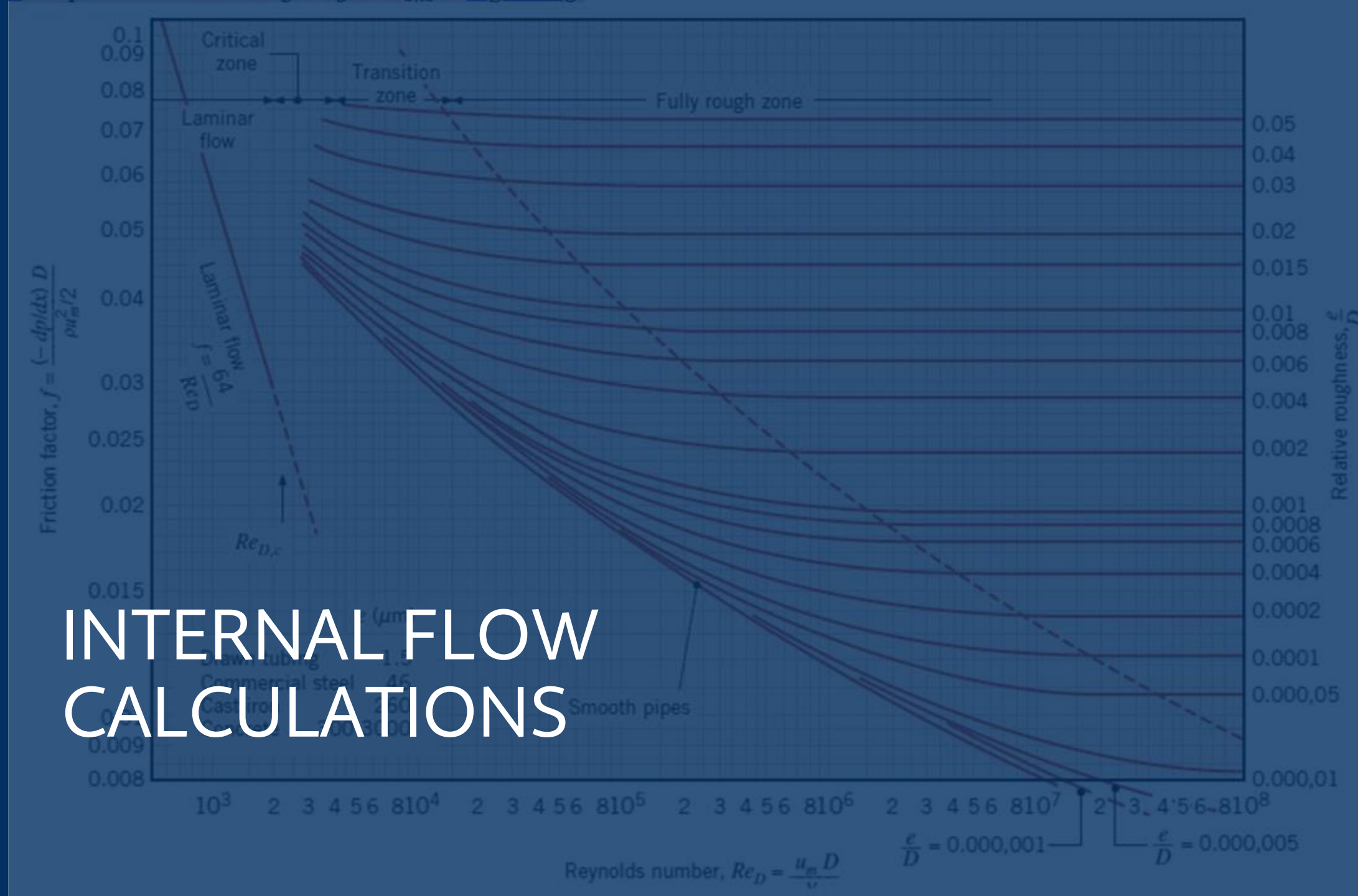
$$\varepsilon \equiv \frac{q}{q_{\max}} \qquad (11.19)$$

$$\mathrm{NTU} = -\frac{\ln[1-\varepsilon(1+C_r)]}{1+C_r} \qquad (11.28b)$$

$$\mathrm{NTU} = \frac{1}{C_r-1}\ln\left(\frac{\varepsilon-1}{\varepsilon C_r-1}\right) \quad (C_r<1) \qquad (11.29b)$$

$$\mathrm{NTU} \equiv \frac{UA}{C_{\min}} \qquad (11.24)$$

- Using Equation 11.19, **ε = 0.464**
- Equation 11.28b for parallel flow: NTU = 1.23
- Equation 11.29b for counter flow: **NTU = 0.852**
- U_{min} = 850 W/m^2 K and U_{max} = 1700 W/m^2 K constraints from Table 11.2 were used with Equation 11.24 to give area minimums and maximumns
- Parallel flow: $A_{s,pf,min}$ = 81.66 m^2, $A_{s,cf,max}$ = 163.32 m^2
- Counter flow: $A_{s,cf,min}$ = 56.58 m^2, $A_{s,cf,max}$ = 113.16 m^2.
- **Counter flow method was chosen**



INTERNAL FLOW CALCULATIONS

PRESSURE AND POWER LOSS

- Final iteration: **n = 38, D = 0.05 m, D_{i2} = 0.05398 m, L = 9 m**
- For loop
- Outer diameter
- Cross-sectional areas: $A_{c,i} = 1.96(10^{-3})$ m^2, A_c = 1.131m^2
- Surface areas: $A_{s,i} = 58.00$ m^2, $A_{s,o} = 33.93$ m^2
- Imaginary diameter
- Equation 8.5 with the mass flow rate divided by n gave V = .383 m/s
- Reynolds number (Eq. 8.6) for inside of tubes and through shell both give turbulent values

$$\dot{m} = \rho u_m A_c \qquad (8.5)$$

$$Re_D = \frac{4\dot{m}}{\pi D \mu} \qquad (8.6)$$

$$Nu_D = 0.023\, Re_D^{4/5} Pr^n \qquad (8.60)$$

$$Nu_D \equiv \frac{hD}{k} \qquad (8.53)$$

$$D_h \equiv \frac{4A_c}{P} \qquad (8.66)$$

$$\frac{1}{UA} = \frac{1}{U_i A_i} = \frac{1}{U_o A_o} \qquad (11.5)$$

$$= \frac{1}{h_i A_i} + \frac{R''_{f,i}}{A_i} + \frac{\ln(D_o/D_i)}{2\pi k L} + \frac{R''_{f,o}}{A_o} + \frac{1}{h_o A_o}$$

PRESSURE AND POWER LOSS

- Hydraulic diameter gave the corrected outer small tube diameter, and was found with Eq. 8.66. Imaginary diameter's cross-sectional area was used to find D_h = 0.05668 m
- Used Nusselt number for inside and outside of tubes to solve for h. Using Eq. 8.60 & 8.53. h_i = 38561.8 and h_o = 2777.7 (Dittus-Boelter)
- Conduction resistance was found to be 3.403(10^{-6}) with Eq. 3.33, and used with the h values to find R_{tot}
- Inverse product of R_{tot} and $A_{s,i}$ gives U ($W/m^2 K$)
- Ideal UA from eq. 11.24 is 9.618(10^4) (W/K)
- The total surface area was found by diving the ideal UA by U, A_s = 80.66 m

$$D_h \equiv \frac{4A_c}{P} \qquad (8.66)$$

$$Nu_D = 0.023\, Re_D^{4/5} Pr^n \qquad (8.60)$$

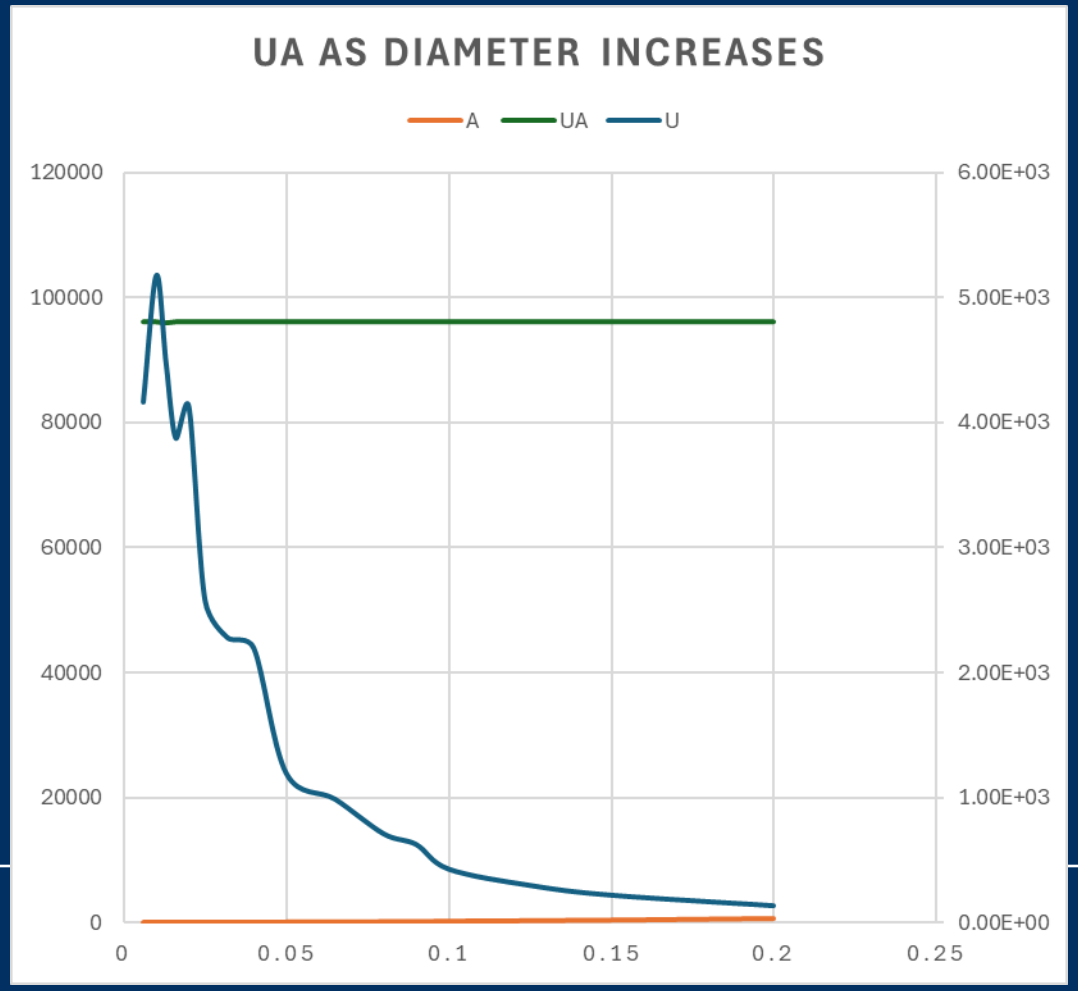
$$Nu_D \equiv \frac{hD}{k} \qquad (8.53)$$

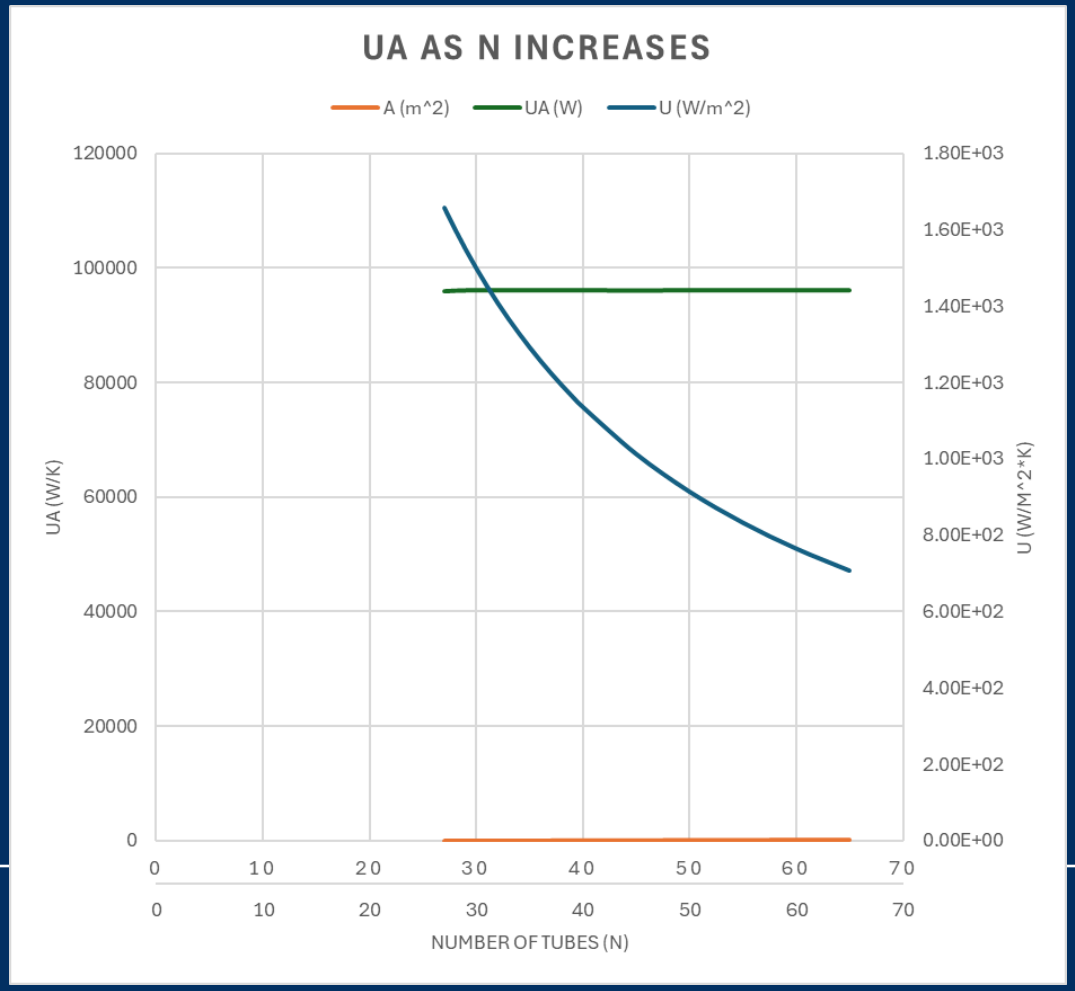
$$R_{t,\text{cond}} = \frac{\ln\,(r_2/r_1)}{2\pi L k} \qquad (3.33)$$

$$\frac{1}{UA} = \frac{1}{U_i A_i} = \frac{1}{U_o A_o} \qquad (11.5)$$

$$= \frac{1}{h_i A_i} + \frac{R''_{f,i}}{A_i} + \frac{\ln(D_o/D_i)}{2\pi k L} + \frac{R''_{f,o}}{A_o} + \frac{1}{h_o A_o}$$

UA VARIATION FROM GEOMETRIC PARAMETERS





PRESSURE AND POWER LOSS

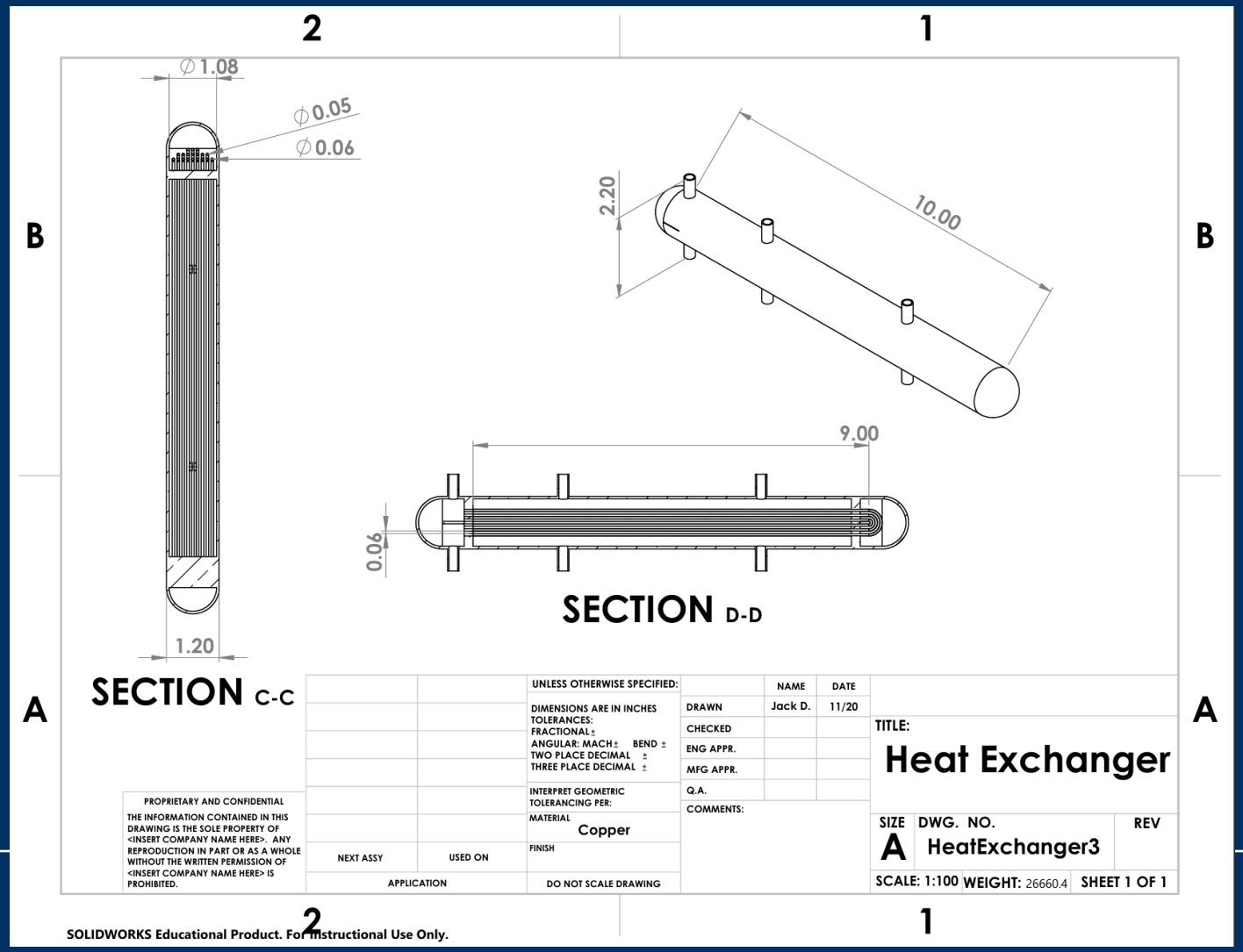
- The roughness was found to be ε/D = 6.000(10^{-4})
- Solving Eq. 8.20, the friction factor *f* = 0.0177
- The pressure drop per pipe was given by Eq. 22a to be ***Δp* = 228.4 pa**, or a total *Δp* of 8.68 kPa
- Eq. 8.22b provides a total power required by the pump, using the total *Δp*, of **P = 239.6 W**, or 6.31 W per pipe

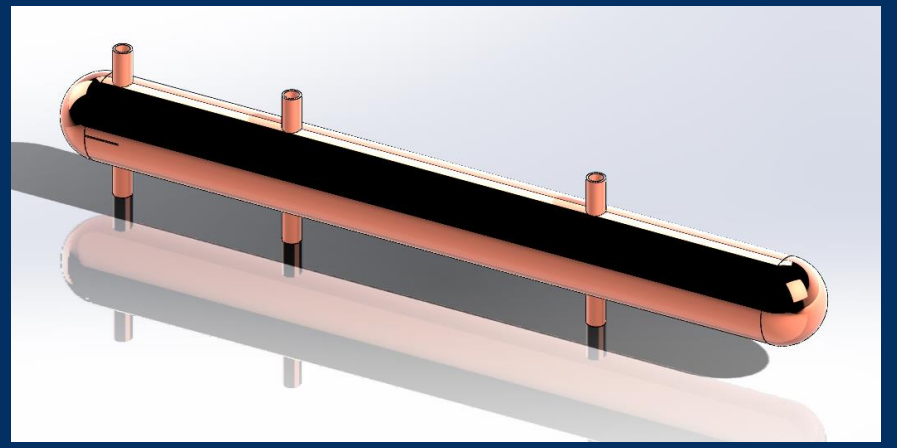
$$\frac{1}{\sqrt{f}} = -2.0 \log \left[\frac{e/D}{3.7} + \frac{2.51}{Re_D \sqrt{f}} \right] \quad (8.20)$$

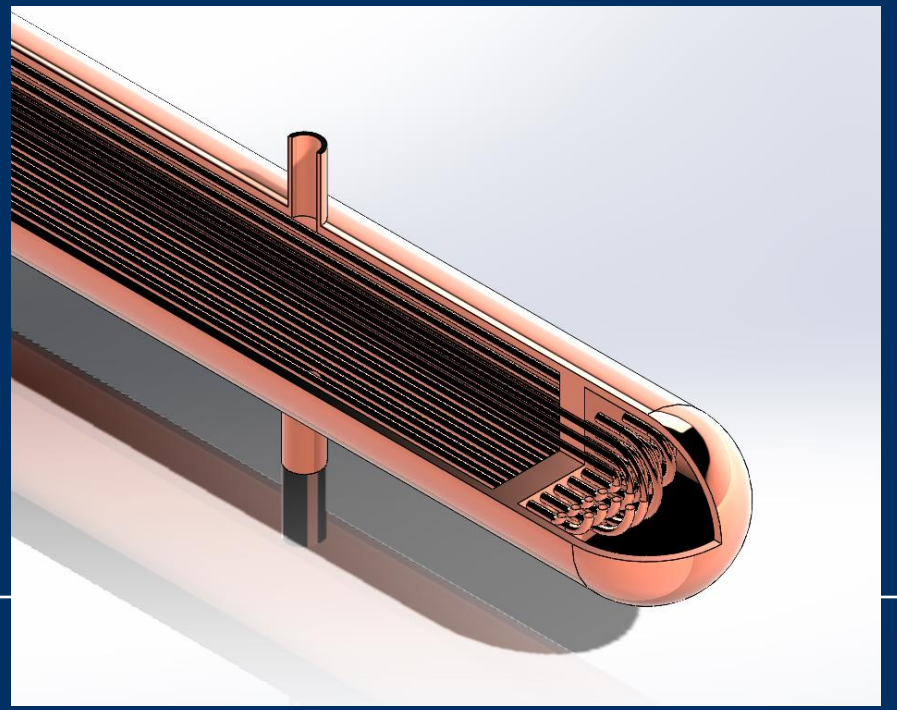
$$\Delta p = -\int_{p_1}^{p_2} dp = f \frac{\rho u_m^2}{2D} \int_{x_1}^{x_2} dx = f \frac{\rho u_m^2}{2D} (x_2 - x_1) \quad (8.22a)$$

$$P = (\Delta p)\dot{\forall} \quad (8.22b)$$

SOLIDWORKS DRAWINGS







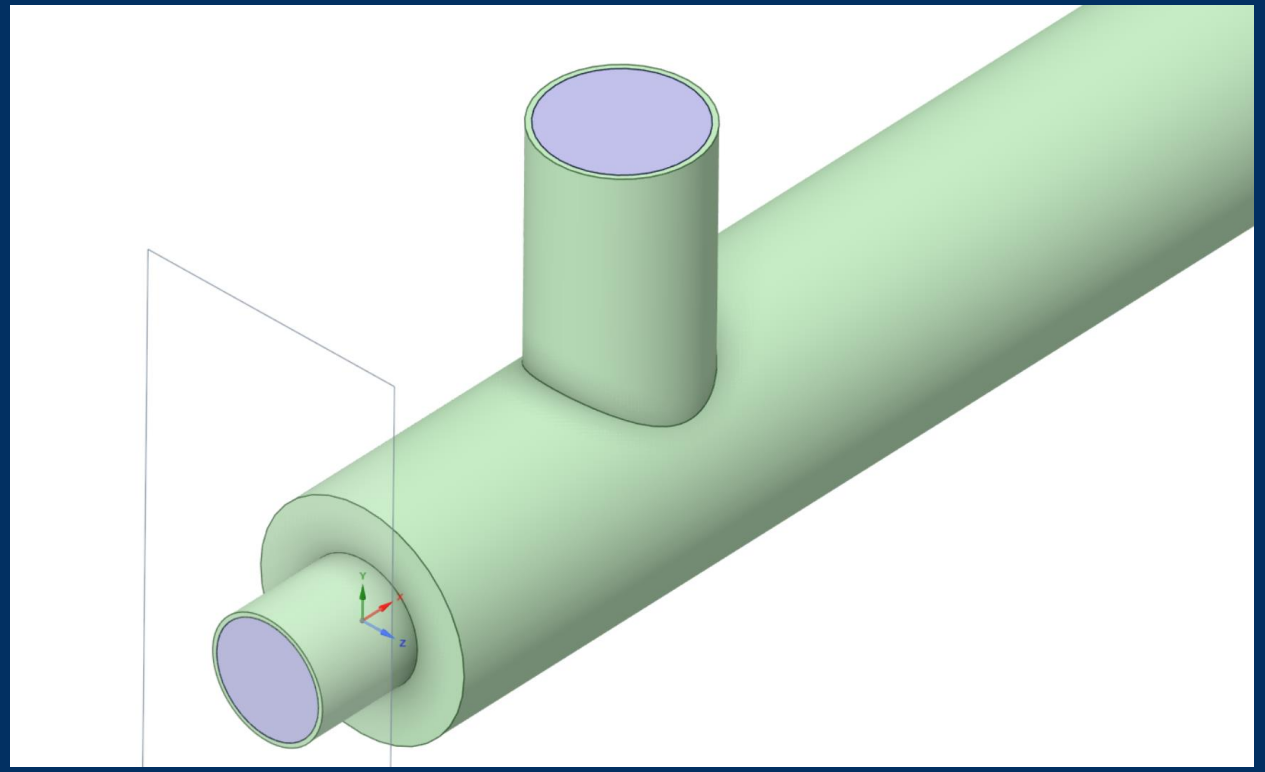
DESIGN JUSTIFICATIONS

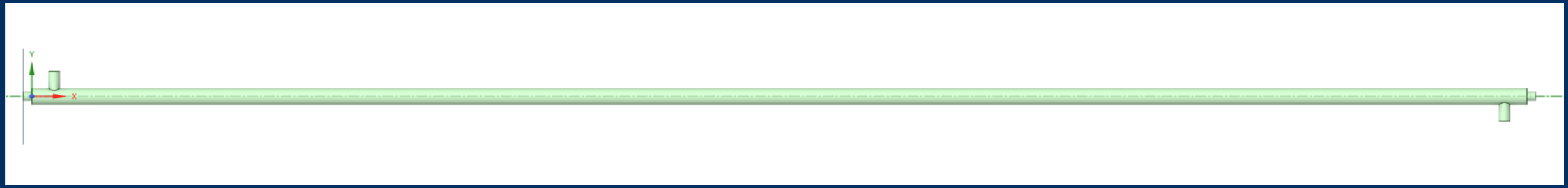
- Effectiveness-NTU vs Internal flow calculations for UA match
- q_{max} = 6.32 MW
- Meets given restraints
  - Length, Umin, Umax
- Geometry iterations maximized the heat transfer coefficient
- Geometry iterations minimized area of total heat exchanged



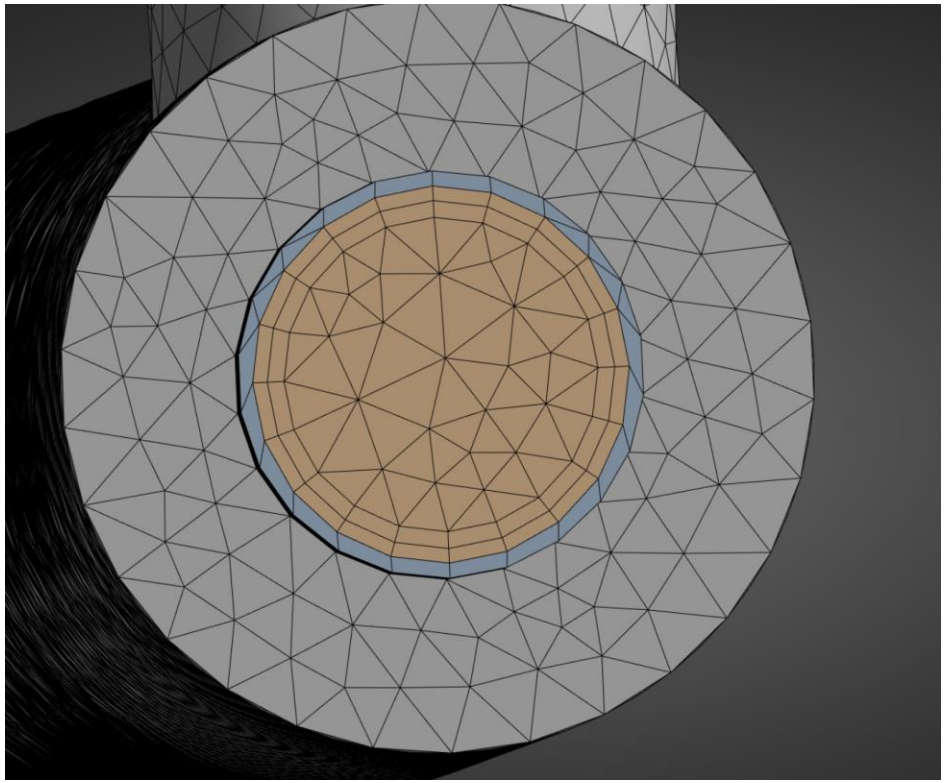
ANSYS SIMULATION

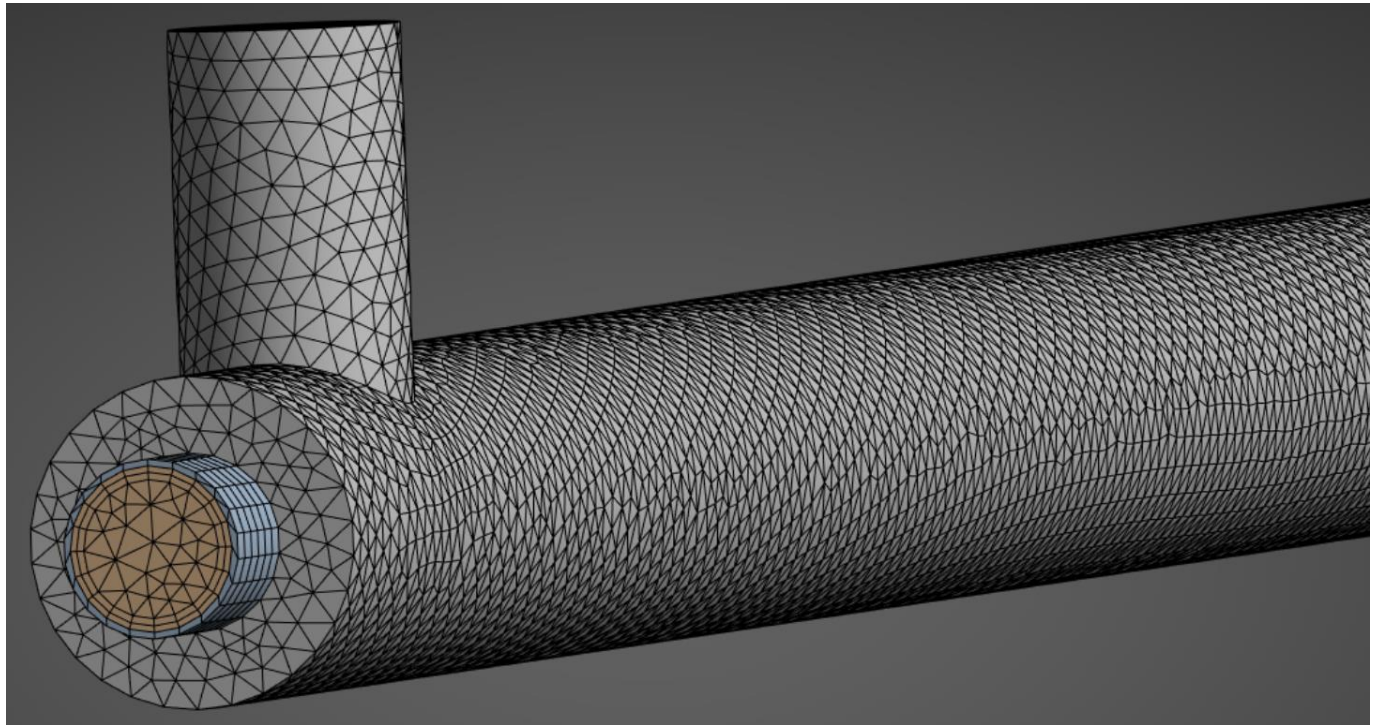
GEOMETRY METHODOLOGY

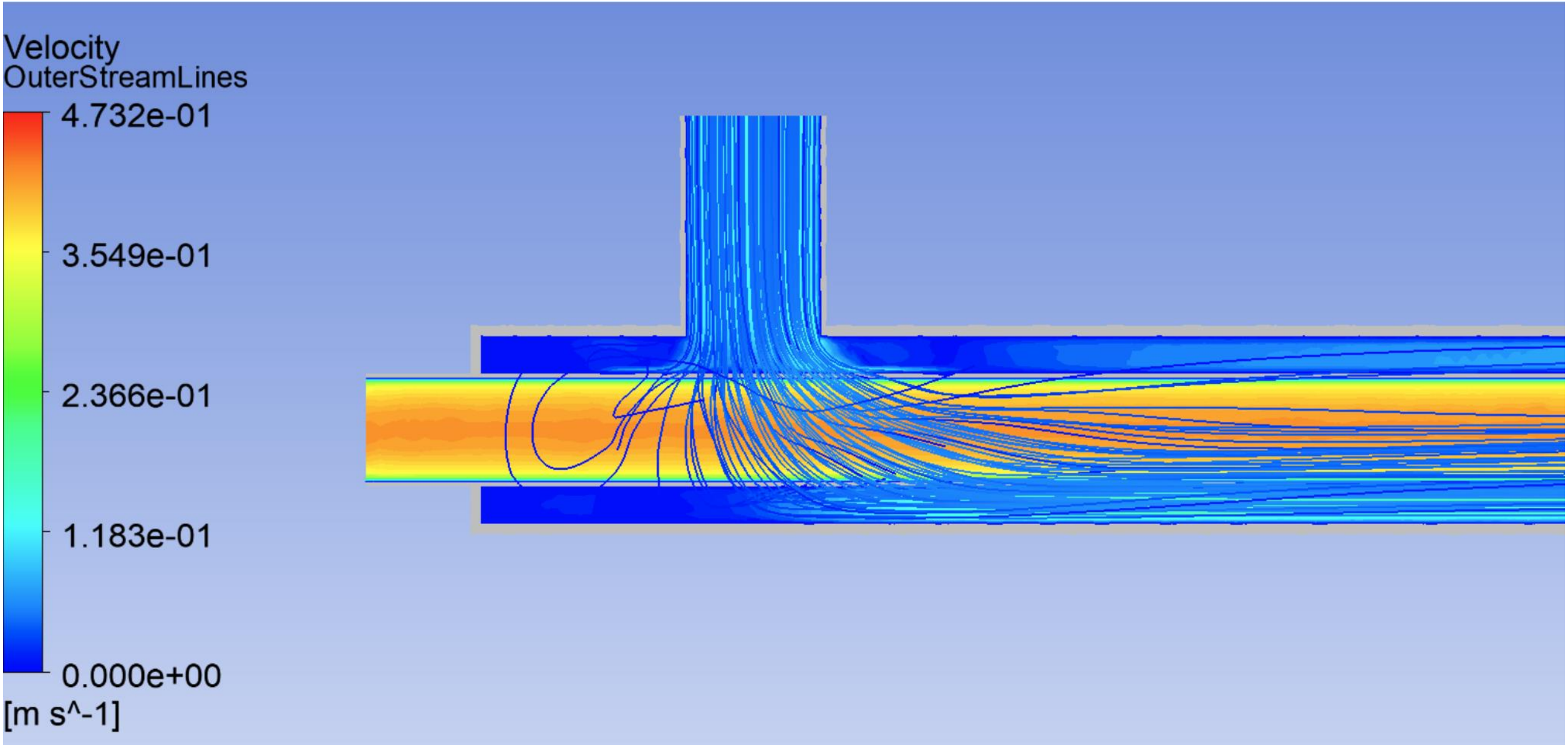




MESH






Velocity
OuterStreamLines
4.732e-01
3.549e-01
2.366e-01
1.183e-01
0.000e+00
[m s^-1]

MESH ITERATION RESULTS

NEW MESH ITERATIONS					
Mesh			Mesh Quantity		Solution
MESH	Cold	Hot	Nodes	Elements / Cells	Converg
Control	N/A	N/A	237440	830223	N/A
Inflation, Smooth Transition	15L, 1.2GR	15L, 1.2GR	1279833	2353902	N/A
Inflation, Smooth Transition	5L, 1.2GR	5L, 1.2GR	578103	1309876	N/A
Inflation, Smooth Transition	3L, 1.2GR	3L, 1.2GR	418824	1112976	N/A
Inflation, Smooth Transition	2L, 1.2GR	2L, 1.2GR	340128	1019518	Yes

Cmin	*(massFlow()@inlet_outer_tube) * 4180 [J kg^-1 K^-1]*
Cr	*Cmin / Cmax*
Current Phase Position	*0*
Current Time Step	*424*
Effectiveness	*Heat Loss Rate / Qmax*
Heat Loss Rate	*(massFlow()@inlet_outer_tube) * 4180 [J kg^-1 K^-1]*(TempMeanOuterOutlet - 307 [K])*
Massflowin	*massFlow()@inlet_inner_tube*
NTU	*1 / (Cr-1) * ln((Effectiveness - 1) / ((Effectiveness * Cr) - 1))*
Phase	*Current Phase Position*
Qmax	*Cmin * (TempMeanInnerInlet - 307 [K])*
Sequence Step	*424*
Shell Mass Flow Rate	*Velocity * Density * .0055655*
TempMeanInnerInlet	*areaAve(Temperature)@inlet_inner_tube*
TempMeanInnerOutlet	*areaAve(Temperature)@outlet_inner_tube*
TempMeanOuterOutlet	*areaAve(Temperature)@outlet_outer_tube*
Time	*0 [s]*
Tube Mass Flow Rate	*Velocity * Density * .0019635*
atstep	*Accumulated Time Step*
ctstep	*Current Time Step*

ANSYS SIMULATION VS. HAND CALCULATIONS

- Heat Rate
- Effectiveness
- Number of Transfer Units

Temperature of the hot water outlet (K, c)	*338.02, 65.02*	*358.73, 85.73*
Maximum heat rate (W/m^2)	*6320160*	*33783.7*
Effectiveness	*0.4643*	*0.584481*
NTU	*0.852*	*0.940509*
Minimum Area of inside tubing (m^2)	*56.58*	*mfhi=.7369*
Maximum Area of inside tubing (m^2)	*113.16*	*mfci=0.14394*

ETHICAL CONSIDERATIONS

- Safety and Reliability
- Enviornmental Impact
- Energy Efficiency



REFERENCES

[1] www.jugaltube.com, “Copper pipe types and colours.”

[2] Copper Tubing Data - Type L | Taylor Walraven.” https://taylorwalraven.ca/copper-tubing-data-Type-L.php

[3] T. L. Bergman, *Fundamentals of heat and mass transfer*. John Wiley & Sons, 2011.

[4] “Gallery of Solved Please Use Graphs and Charts Provided to Solve Pro - Pipe Roughness Chart | Values of Relative Roughness for Pipes Made of Common, Solved Show All Work Please Will Rate F 316 Re 25 F, Chiller Piping Systems Edu 2015, Solved a 500 M Long Galvanized Iron Pipe Has a 1 2 M Diam, Moody Diagram.” https://keski.condesan-ecoandes.org/pipe-roughness-chart/solved-please-use-graphs-and-charts-provided-to-solve-pro.html